Quest of Aivengarde: Comparative Study of Player Experience Across LLM Dialogue Systems

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Figure 1: In-game scene from Quest of Aivengarde, showing the player character beside the goblin camp.

Abstract

Non-player character (NPC) dialogue plays a crucial role in games. Narrative-driven video games in particular depend on NPCs to help shape the player experience by contributing to narrative, immersion, and player agency. Current technologies allow large language models (LLMs) to create dynamic, context-sensitive dialogue for NPCs, yet their impact on player experience remains underexplored. Quest of Aivengarde is a custom-built role-playing game developed as a research testbed for comparing four dialogue system designs: a static control version and three LLM-driven variants that rephrase, hybridize, or fully generate NPC dialogue. Building on previous pilot and demo studies, this paper integrates the system design and full empirical evaluation to examine how levels of generative agency affect interaction quality, immersion, and player engagement. Using mixed-methods analysis of behavioral logs and post-game surveys from 64 participants, results suggest that fully open-ended LLM dialogue fosters longer, more natural conversations and is particularly engaging for casual players. We discuss design trade-offs between narrative control and conversational freedom and propose a practical framework for selecting and combining LLM-driven dialogue approaches in future game development and research.

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CCS Concepts

• **Software and its engineering** → **Interactive games**; • **Human-centered computing** → **User studies**; **Text input**.

Keywords

Large Language Models, NPC Dialogue, Player Experience, Generative AI, Procedural Narrative, Role-Playing Games, Game Design

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1 Introduction

Dialogue systems have long shaped how players experience narrative, character, and choice in games. From early experiments in interactive drama such as *Façade* [20] and *Chris Crawford on Interactive Storytelling* [7] to contemporary procedural dialogue frameworks [21, 31], the design of meaningful player–NPC communication has been central to digital storytelling. Recent advances in large language models (LLMs) extend this tradition by enabling NPCs to generate contextually relevant and adaptive dialogue in real time. Studies and surveys have begun to outline the potential of LLMs in game contexts [11, 30, 37], yet the empirical understanding of how generative dialogue impacts player experience remains limited.

Existing explorations of LLM-driven interaction - such as *AI Dungeon*, *CALYPSO* [40], and co-creative systems like *CoAuthor* [17] - demonstrate technical feasibility but are often evaluated through anecdotal or open-ended play. More recent empirical studies have

addressed this gap through smaller-scale comparisons [2, 13, 23], though these remain fragmented and seldom connect player perception to specific dialogue design choices. As generative models like GPT-5 [1] and Gemini [8] continue to improve, there is a growing need to move beyond feasibility studies toward systematic design and evaluation frameworks.

Quest of Aivengarde was developed to address this need. It is a custom-built role-playing game (RPG) created as a controlled research environment for evaluating dialogue systems under identical narrative and gameplay conditions. The game includes four dialogue versions, ranging from a traditional fixed tree to fully generative open-ended conversation. Earlier work introduced the technical foundation and comparative design framework of this system [28], followed by a pilot study that examined three LLM-driven variants and their effect on player engagement and immersion [24]. Results from the pilot suggested that fully open-ended dialogue fostered longer and more natural interactions, particularly among casual players, while hybrid systems offered a promising balance between structure and freedom but exhibited technical inconsistencies.

Building on these prior studies and the foundation established in the corresponding master's thesis, this paper consolidates, extends, and deepens the analysis of the Quest of Aivengarde project. Rather than presenting new data, it offers a comprehensive synthesis of the full mixed-methods evaluation, integrating quantitative findings, thematic analysis, and design reflection to derive a structured understanding of generative dialogue design. The aim is to move beyond reporting feasibility or isolated results and toward articulating the broader implications of LLM-driven NPCs for narrative game design, authorial control, and player agency.

Specifically, this paper addresses the following research questions:

- **RQ1:** How do different LLM-driven dialogue systems affect player engagement and immersion in narrative games?
- **RQ2:** What design trade-offs emerge between authorial control, system stability, and player agency in generative dialogue design?

Through this integrated perspective, the paper contributes:

- A consolidated account of the design, methodology, and empirical findings from the Quest of Aivengarde project.
- An in-depth interpretation of how levels of generative freedom shape player experience and narrative perception.
- A design-oriented framework that formalizes insights for implementing and evaluating LLM-based dialogue in future games and research.

2 Related Work

2.1 Dialogue Systems in Games

NPC dialogue has traditionally relied on branching trees and pre-authored scripts [7, 9, 31]. While effective for ensuring narrative consistency, these systems limit expressiveness and require extensive authoring. Procedural systems have sought to expand variation through grammars or dialogue planning frameworks [26, 27], but remain bounded by designer-authored rules. Earlier experiments in interactive drama [20] and narrative simulation highlighted the challenges of balancing authorial control with player responsiveness.

2.2 LLM-Based Generative Dialogue

LLMs have introduced new opportunities for expressive, context-sensitive NPCs. Early systems like *AI Dungeon* explored open-ended play [12], while more recent studies have investigated prompt tuning, dialogue safety, and grounding in narrative goals [2, 23, 39]. Surveys by Gallotta et al. [11] and Yang et al. [37] categorize the design space of LLM-integrated games, including concerns over hallucinations [15] and narrative drift [22].

Recent work has also examined collaboration [10], emotional expressivity [38], and character-consistent generation [19]. Systems like LIGS [14] and Slice of Life [32] integrate LLMs into sandbox or emergent environments, while frameworks such as Song's cross-platform architecture [29] and Poglitsch et al.'s agent-based evaluation using *Werewolf* [25] show increasing interest in scalable comparative setups.

2.3 Controllability and Prompt Design

Prompt engineering, constraint templates, and function-calling interfaces are increasingly used to steer LLM output for narrative control [3, 5, 34, 36]. Techniques like symbolic grounding [32] or dialogue discourse parsing [18] help mitigate generative unpredictability. Whitehead et al. [35] investigate conversational interfaces to drive procedural systems, showing how authorial goals can co-exist with naturalistic language.

2.4 Player Agency and Narrative Experience

Several studies have addressed how dynamic dialogue affects immersion and agency. Role-playing experience is shaped by believability, reactivity, and consistency of NPC behavior [21, 26, 33]. Akoury et al. [2] explored how players perceive generative dialogue in commercial games, while Taylor-Giles [31] emphasized user-centered design for branching systems. Emotional engagement and empathy with NPCs are also active areas of exploration [38].

2.5 Positioning This Work

While prior research has demonstrated the feasibility and creative potential of LLM-driven NPCs, few studies offer empirical comparisons of multiple dialogue architectures in identical gameplay conditions. Quest of Aivengarde extends this landscape by combining fixed, hybrid, and open-ended dialogue in a unified RPG environment. Building on previous design and pilot work [24, 28], this paper provides a systematic evaluation of how dialogue system design influences player engagement, perception, and interaction quality.

3 System Design: Quest of Aivengarde

Quest of Aivengarde is a custom-built third-person role playing game that serves as a controlled research testbed rather than a commercial prototype (See Figure 7). The game is set in a small medieval town whose food stores have been stolen by goblins. Players must speak with NPCs to uncover information that helps them complete a sequence of challenges, such as choosing the safe path through a forest or learning the magic word that breaks a

barrier. All visual assets, mechanics, and quest logic are identical across conditions, which allows the study to isolate the impact of the dialogue system on player experience.

The town contains a small cast of key NPCs that each gate specific pieces of information. The Bard introduces the story and sends the player to the Innkeeper, who provides a world map and a notebook that records discovered clues. Other central NPCs include the Blacksmith, Herbalist, Thief, Huntsman, Wizard, and a Hobgoblin guarding the stolen food. Each is linked to a specific challenge and is authored with a distinct personality, rhetorical style, and knowledge scope. Regardless of version, the same knowledge is available from the same NPCs and the same challenges must be solved in the same order. This shared structure ensures that differences in behavior can be attributed to the dialogue systems rather than to changes in content or progression.

3.1 Dialogue System Variants

The core independent variable in Quest of Aivengarde is the dialogue system. Four versions span a spectrum from fully authored dialogue to fully generative interaction:

Control Version (fixed tree). The *Control Version* uses a traditional branching dialogue tree. Each node contains one NPC line and a small set of fixed player replies, and selecting a reply deterministically advances the conversation to the next node. All text and branching structures are manually authored, with no LLM involvement. This version provides a stable baseline that reflects common practice in narrative RPGs. An overview of the structure is shown in Figure 2.

Version A (LLM paraphrasing). Version A keeps the same underlying dialogue tree as the *Control Version*, but routes each authored node through a Rephraser Module before it is presented to the player. At the start of a conversation, all NPC lines and associated player options for that interaction are sent to an LLM with instructions to rephrase the text while preserving meaning, required information, and character tone. The regenerated lines replace the original authored text, introducing linguistic variation across playthroughs while leaving branch structure and quest logic unchanged. Figure 3 illustrates the process.

Version B (hybrid fixed plus open input). Version B extends the fixed tree with an additional option to write free-text to the NPC. Players may either select an authored response or type a message. free-text is sent to an NPC Module that receives the current node, the full dialogue tree, and a character profile describing the NPC's role, personality, and constraints. The module generates an in-character reply and a small set of follow-up options that connect back into the authored structure. New dialogue nodes are dynamically inserted so that open-ended branches can rejoin existing authored nodes or terminate in controlled exits, preserving narrative progression and quest logic. A schematic of the interaction flow is shown in Figure 4, while Figure 6 illustrates how these hybrid nodes are created and integrated into the dialogue tree.

Version C (fully open ended dialogue). Version C removes the visible dialogue tree entirely and relies on fully open-ended text interaction. Players always type their own messages, which are passed to the NPC Module together with a concise summary of quest state and the NPC's character profile. The module generates one response at a time and maintains an NPC-specific conversation history so it can reference earlier turns. Quest progression and the release of mandatory information are still governed by separate control modules, but the moment-to-moment conversation is entirely shaped by the player and the LLM. Figure 5 shows the high-level structure of this system.

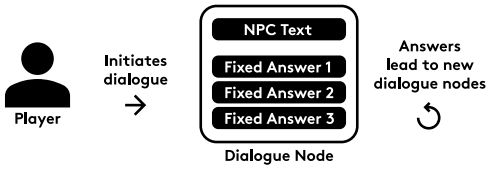


Figure 2: Structure of the fixed branching dialogue system in the *Control Version*. Reproduced from [24]

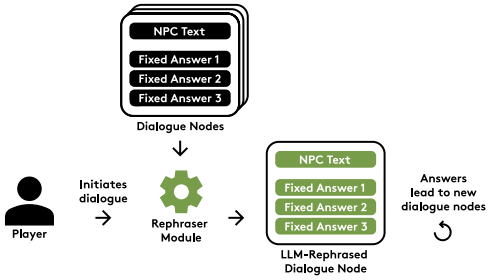


Figure 3: Dialogue flow in *Version A*, where authored text is paraphrased by an LLM before being shown to the player. Reproduced from [24]

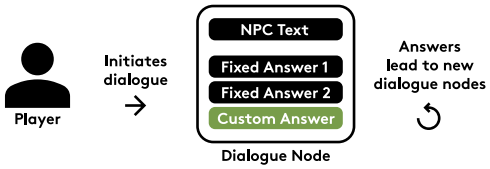


Figure 4: Hybrid interaction in *Version B*, where players can choose fixed replies or enter free-text that is processed by the NPC Module. Reproduced from [24]

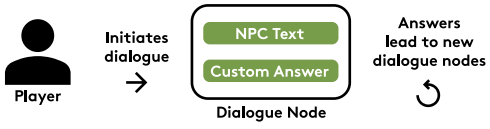


Figure 5: Fully open-ended dialogue system in *Version C*, where all player input is handled by an LLM with NPC-specific memory and character context. Reproduced from [24]

3.2 LLM Integration and Architecture

All versions were implemented in Unity with C# scripts that communicate with OpenAI's GPT-4o model via HTTP requests. Rather than relying on a single monolithic model, the system uses a modular multi agent design with separate LLM instances for distinct responsibilities:

- **NPC Module** (Versions B and C) generates natural language replies on behalf of a specific NPC. Each instance maintains its own dialogue history and uses a two part prompt that combines a character sheet (role, backstory, speaking style, and constraints) with a dialogue system prompt that explains how to respond to the player and when to defer to game logic. Conversation histories are trimmed to the most recent turns to respect token limits and reduce drift.
- **Function Call Module** (Versions B and C) inspects player input for utterances that should trigger in game actions, such as revealing a critical hint, updating the notebook, or ending a conversation. Prompts instruct the LLM to return a lightweight JSON specification of which function to call and with which arguments. The Unity side validates and executes the call without exposing internal game state directly to the NPC Module.
- **Quest Completion Module** (Versions B and C) checks whether the player's message satisfies predefined completion or failure criteria for the current challenge. It receives the message together with a compact description of the active quest objectives and returns a categorical verdict (`Completed`, `Failed`, or `None`), which the quest manager uses to update the game state.
- **Rephraser Module** (Version A) takes authored dialogue lines as input and returns rephrased variants packaged as a structured JSON object, which is then parsed back into the game's dialogue format.

Each module is invoked separately and only receives the minimum context required for its task. This separation of concerns keeps prompts concise, limits cross contamination between NPCs, and allows individual modules to fail without compromising the entire system. It also makes it possible to reuse the same NPC and quest logic across Versions A, B, and C while varying only how the underlying LLMs are used.

This design supports the study's comparative goals. All four dialogue systems operate in the same world with the same NPCs, challenges, and quest progression, which provides experimental control. At the same time, the modular LLM architecture allows the prototype to explore a wide range of interaction styles, from deterministic tree based dialogue to fully open ended conversation, without sacrificing narrative consistency or technical stability.

Free-text input and response control. In Versions B and C, players were able to enter free-text messages when interacting with all NPCs. No explicit lexical filtering or predefined input constraints were imposed beyond standard input sanitation. Instead, relevance and narrative grounding were encouraged through prompt design and system architecture. NPC responses were generated using character-specific prompts that constrained role, knowledge scope, and narrative context, while critical gameplay logic was handled by separate modules. Player input was independently evaluated by the Function Call and Quest Completion modules to determine whether it triggered quest-related actions or progression, ensuring that game state changes were decoupled from the surface-level conversational response.

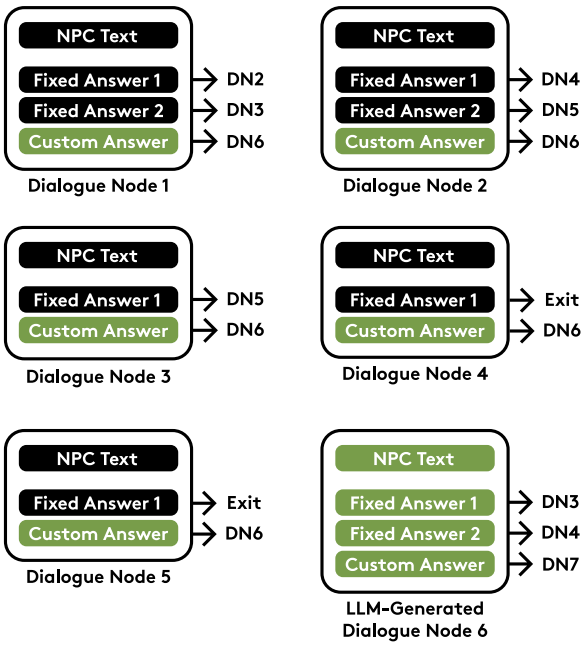


Figure 6: Depicts how new LLM-generated dialogue nodes are added to the tree structure, preserving the conversation flow while dynamically expanding possible interactions.



Figure 7: The top picture shows a bird's-eye view of the town. The bottom picture shows the player character (left) and the Innkeeper (right) inside the town inn. Reproduced from [24]

4 Methodology

4.1 Study Design

The study employed a mixed-methods design to evaluate how different LLM-driven dialogue systems affect player engagement, immersion, and perception of agency in narrative games. The experiment was conducted within Quest of Aivengarde, a custom-built RPG designed as a controlled testbed where all narrative, gameplay, and aesthetic elements remain constant across dialogue versions. Each participant played one of four versions of the game, corresponding to a distinct dialogue system: *Control* (fixed tree), *Rephrase* (LLM paraphrasing), *Hybrid* (fixed + open-ended), and *Open-Ended* (fully generative dialogue).

Building on the design and pilot studies [24, 28], this paper re-examines the same experimental dataset through an expanded analytical lens. Rather than introducing new data collection, the focus lies on integrating the complete behavioral logs, survey responses, and qualitative feedback into a comprehensive mixed-methods analysis. This approach enables a deeper understanding of how dialogue design choices shape player engagement, immersion, and perceived agency across the four versions of Quest of Aivengarde.

4.2 Participants and Procedure

Participants were recruited through university channels and online communities, targeting individuals familiar with computer games and proficient in English. This ensured a participant pool representative of typical RPG players. Prior to play, participants were briefed on the study's purpose, data privacy, and withdrawal rights, followed by informed consent in accordance with institutional ethics guidelines.

Each session followed a three-part protocol:

(1) **Introduction:** Participants received a short explanation of the controls, objectives, and purpose of the test, without disclosing specific study hypotheses.
(2) **Playthrough:** Each participant was assigned a version of the game using a stratified randomization algorithm that balanced the number of sessions per dialogue condition.
(3) **Questionnaire:** After completing the game, participants filled out a structured post-game survey prefilled with run metadata such as version and unique run code.

A total of 64 participants completed the study, divided evenly across the four dialogue systems (see Table 1). Each play session lasted between 10–30 minutes, depending on dialogue length and interaction choices.

Demographic Variable	Value
Total Participants	64 (62 analyzed)
Mean Age (SD)	25.2 (4.57)
Gender	77.4% Male, 22.6% Female
Dyslexia Reported	11.3%
Color Blindness Reported	3.2%
Participants per Version	Control: 16, A: 15, B: 16, C: 15

Table 1: Participant demographics summary

4.3 Data Collection

Two complementary data sources were used to capture both behavioral and subjective dimensions of player experience.

4.3.1 Game Data. All gameplay data were logged automatically to one of four NoSQL databases, one per dialogue version. Each database contained seven structured collections:

- **Dialogue Messages:** Complete logs of NPC and player utterances with timestamps, enabling quantitative and qualitative dialogue analysis.
- **Dialogue Time:** Duration of active dialogue per interaction and per NPC.
- **Movement Data:** Player coordinates sampled every 0.25 seconds to analyze exploration patterns and pacing.
- **Area Time:** Total time spent in each region of the game world.
- **Area Transitions:** Counts and sequences of area changes, identifying common navigation routes.
- **Other Metrics:** Total run duration, quest completions and failures, LLM response times, notebook use, and in *Version B*, the frequency of static vs. open-ended dialogue selections.
- **Run Codes:** Unique alphanumeric identifiers linking gameplay data to the corresponding questionnaire responses.

4.3.2 Questionnaire Data. Upon completion, participants answered a post-game questionnaire consisting of 24 Likert-scale items and four open-ended questions. The survey covered nine thematic sections: consent, background information, general experience, dialogue system evaluation, clarity and engagement, AI impressions, quest difficulty, and free-form feedback. This structure supported quantitative comparison and thematic coding of qualitative insights.

4.4 Data Analysis

Quantitative data from gameplay metrics and questionnaires were analyzed using descriptive statistics and inferential tests to compare performance and perception across dialogue versions. One-way ANOVA was applied to continuous variables, while non-parametric Kruskal–Wallis tests were used for ordinal data. Significant results ($p < .05$) were followed by Tukey's HSD or Wilcoxon post-hoc tests, with p-values adjusted for multiple comparisons. Effect sizes were reported using eta squared (η^2).

Key dependent variables included total playtime, dialogue time per NPC, number of interactions, quest outcomes, LLM response delays, and engagement ratings. Additional analyses segmented participants by player type, categorized as *casual* (fewer than 10 hours of gameplay per week) or *hardcore* (10 hours or more per week), to explore potential interaction effects.

Qualitative feedback was analyzed using inductive thematic coding to identify patterns in player perception of naturalness, control, and immersion. Dialogue transcripts were sampled for recurring conversational behaviors, misunderstandings, and emergent narrative expressions. Findings from both data streams were triangulated to contextualize the quantitative outcomes and extract design insights.

Dialogue Version	Casual	Hardcore
Control Version	12	4
Version A	7	9
Version B	10	5
Version C	7	9
Total	**35**	**27**

Table 2: Distribution of casual and hardcore players across dialogue versions. Casual players report fewer than 10 hours of gameplay per week, while hardcore players report 10 hours or more.

4.5 Ethical Considerations

All procedures adhered to standard research ethics. Participants' data were anonymized through randomly generated run codes and stored in compliance with institutional data protection regulations. Participants could request data deletion at any time.

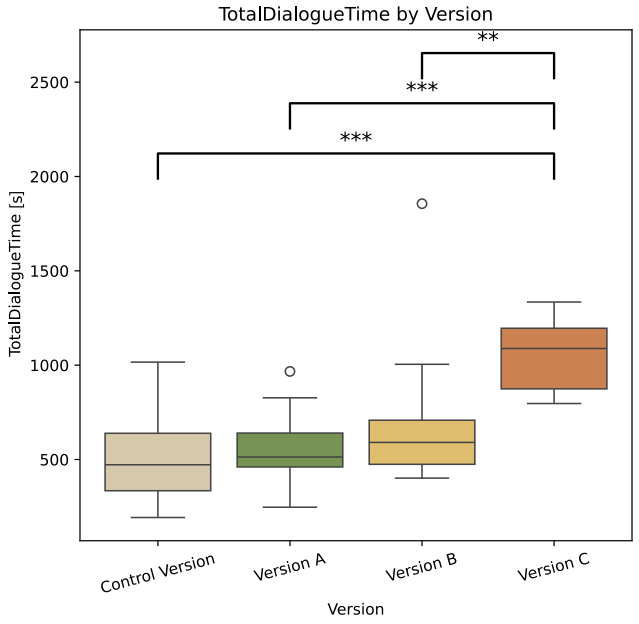


Figure 8: Total dialogue time (in seconds) by game version.

5 Results

5.1 Quantitative Results

Across all metrics, *Version C* consistently produced longer gameplay and dialogue times (both measured in seconds) compared to the other systems. As shown in Figure 8, a Kruskal–Wallis test revealed statistically significant differences in both total gameplay time ($p = 0.0064$, $\eta^2 = 0.152$) and dialogue time ($p < .001$, $\eta^2 = 0.392$), with post-hoc tests confirming *Version C*'s distinction across pairwise comparisons. Interestingly, while *Version B* led to longer dialogue exchanges per interaction, *Version C* accumulated more overall dialogue time.

When dialogue time was subtracted from total session duration, no significant differences remained between versions, indicating that the increased playtime in *Version C* was driven primarily by extended conversational engagement rather than differences in non-dialogue gameplay.

The number of interactions varied modestly across systems. Players in the *Control Version* initiated significantly more interactions than those in *Version A* ($p = 0.0102$), while *Version B* encouraged longer turn-based exchanges once conversations began (see Figure 9). No significant differences were found between casual (< 10 hours/week) and hardcore (≥ 10 hours/week) players in terms of total playtime or dialogue engagement.

Analyzing individual NPC conversations revealed that *Version C* resulted in significantly longer interactions for key NPCs, especially the Blacksmith, Hobgoblin, and Wizard (all $p < .001$). This pattern reflects the expansive affordances of the open-ended system. Out-of-dialogue time, however, did not differ across versions ($p = 0.3739$), suggesting that core gameplay outside conversations remained consistent.

Quest-level analysis showed significant time differences only for quest: *Break the Barrier* ($p < 0.001$, $\eta^2 = 0.234$), again implicating *Version C* due to its increased exploratory and conversational scope. Quest failure rates showed no statistical difference.

Loading times also varied notably across systems. *Version C* offered the fastest single-turn response times ($p < 0.001$), yet it accumulated the highest total LLM delay across the full playthrough ($p < 0.001$). This pattern reflects the much larger volume of dialogue generated in *Version C*, rather than slower model performance.

Survey data highlighted positive perceptions of *Version C* and *Version B*, particularly regarding engagement, enjoyment, and freedom. Three survey items showed significant overall differences: interest in talking to NPCs ($p = 0.0289$), enjoyment of the dialogue system ($p = 0.0402$), and perceived unnatural dialogue moments ($p = 0.0207$). Post-hoc analyses identified *Version C* as outperforming *Version A* in NPC interest ($p = 0.0483$).

5.2 Qualitative Results

Positive Feedback. Participants complimented *Version C* for its freedom, immersion, and personality-rich NPC responses. *Version B* was appreciated for its balance of free input and structured dialogue, offering fluid yet grounded interaction. *Version A* and the *Control Version* were seen as coherent and clear, though less flexible.

Negative Feedback. Across all versions, limitations in agency and responsiveness were the most common issues. In *Version C*, these often stemmed from misunderstanding or lack of follow-through by the model. Dialogue inconsistency and UI challenges were also noted, particularly in *Version B*.

Suggestions. Players frequently requested improvements to UI navigation (such as maps or hotkeys), better dialogue memory, and expanded story depth. In *Version C*, some participants sought clearer gameplay feedback and more consistently responsive NPC behavior.

5.3 Player Input Behavior

Manual review of free-text responses revealed high levels of in-character, grounded interaction. In *Version C*, over 95% of open-ended responses respected the narrative setting, with few examples of off-topic or immersion-breaking content. *Version B* showed slightly more variation, though most free-text entries enriched the role-play rather than disrupted it.

5.4 Key Takeaways

The results indicate that open-ended LLM dialogue systems substantially increase player engagement and immersion, especially among casual players. However, they also introduce challenges in stability, responsiveness, and total system load. Hybrid systems like *Version B* can offer a workable compromise by maintaining structured narrative control while allowing expressive input, though they require careful design to avoid mechanical inconsistencies.

6 Discussion

6.1 Interpreting The Results

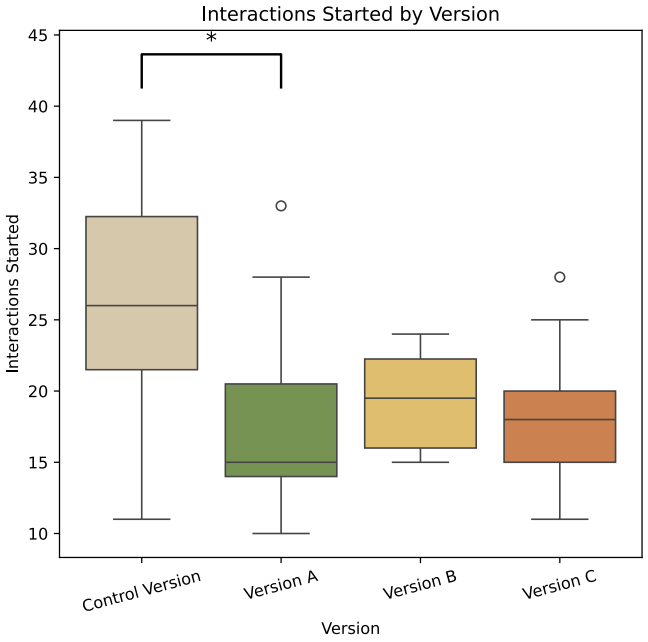
The results strongly suggest that the degree of generative freedom that the system affords directly correlates with player immersion and engagement. The fully open-ended system of *Version C* produced significantly longer and more extensive conversations compared to the other versions, aligning with the idea that the freedom to use natural language fosters a deeper sense of agency and role-play. This is further evidenced by the highest total dialogue time and the strongest positive subjective feedback regarding interest in talking to NPCs and enjoyment of the dialogue system. The hybrid system of *Version B* also scored well, demonstrating that the balance between fixed and free structure can enhance the experience. In contrast, *Version A*, while more predictable, provided limited expressive agency, which reduced the capacity for players to enhance their engagement beyond the *Control Version*.

6.2 Study and System Limitations

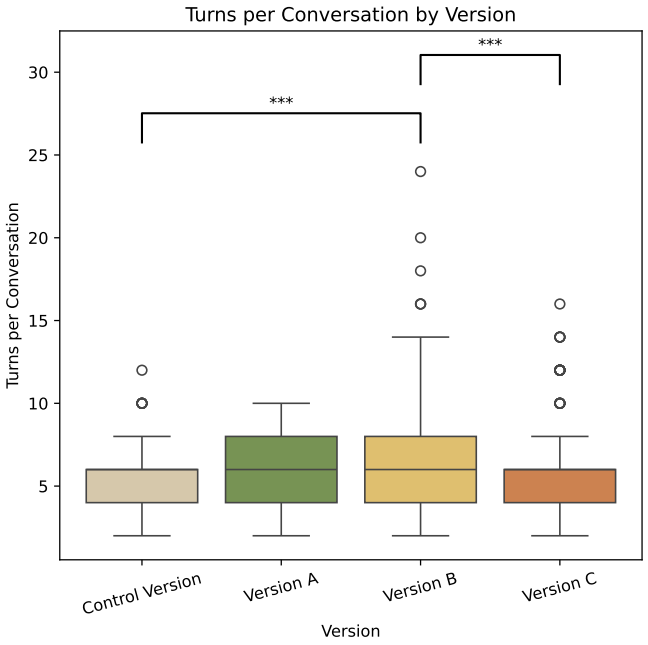
Several limitations must be acknowledged when interpreting the findings of this study. While the results provide valuable insights into how different LLM-driven dialogue systems shape player interaction, constraints related to system performance, narrative coherence, study design, and generalisability remain.

Technical and System Constraints. The modular, multi-agent LLM architecture was designed to enhance stability and reliability. Despite this, technical limitations remained a practical concern. Although *Version C* exhibited the fastest single-turn response times among the LLM-driven variants, it accumulated the highest total LLM delay due to the substantially larger volume of dialogue generated. Participant feedback indicated that even modest increases in response latency negatively affected trust in and immersion with NPCs. Qualitative observations further revealed occasional hallucinations, resulting in inconsistent or misleading information.

Version B introduced additional challenges related to its hybrid design. The integration of generative dialogue within a fixed narrative structure often produced coherence issues, as the LLM was heavily constrained by predefined dialogue paths. This frequently prevented logical or emotionally consistent follow-ups to player input, leading to narrative irregularities and lower coherence scores compared to the fully open-ended approach of *Version C*.



(a) Interactions started by game version.



(b) Response count per game version.

Figure 9: Comparison of interactions started and total response count across all game versions.

Prototype and Narrative Design Limitations. The game was developed as a research prototype rather than a polished commercial product. While preliminary usability testing was conducted, time and resource constraints resulted in minor bugs, unrefined feedback systems, and visual inconsistencies. Although no critical failures occurred, these factors may have influenced immersion and, by extension, player behaviour and subjective evaluations.

The prototype format also limited narrative depth. A more content-rich and narratively dense experience could encourage longer and more meaningful NPC interactions, potentially revealing different engagement patterns and more fully demonstrating the affordances of LLM-driven dialogue systems.

Sample Size and Player Demographics. The study included 64 participants distributed across four game versions, with 15–16 participants per condition. While sufficient for an exploratory pilot study, each group approaches the lower bound for detecting statistically significant effects, limiting the power to identify more subtle differences between versions [6]. Increasing the sample size would improve reliability and support more robust comparisons.

Participants primarily shared similar levels of gaming literacy and familiarity with standard interaction mechanics. This reduced confounding usability issues but constrained external validity, as interaction styles and preferences may differ in more heterogeneous player populations.

Design Scope and Generalisability. A between-subject design was employed to avoid carryover effects and memory bias [4, 16]. However, this prevented participants from directly comparing dialogue systems, requiring subjective evaluations to be interpreted within the context of single-version exposure.

Finally, the fixed narrative structure, predefined quest logic, and specific interaction design limit the generalisability of the findings. While the observed design implications may inform future work, further studies are required to examine how similar LLM-driven dialogue systems perform across different genres, narrative structures, and interactive contexts.

6.3 Trade-offs

Across the measures of dialogue length, total playtime, and subjective reports during gameplay, *Version C* consistently produced a higher level of conversational play. Players in *Version C* appeared to treat NPCs less as quest dispensers and more as conversational partners. However, this freedom also introduces conversational misunderstandings, with several instances of ambiguity in the system's responses. *Control Version* and *Version A* ensured reliable narrative progression but provided limited expressive agency. Conversely, *Version C* enabled rich role-play at the cost of system reliability and narrative consistency. The hybrid approach of *Version B* offered a middle ground but introduced interface-level ambiguity for several players. These differences raise concerns when it comes to the trade-offs between authorial control and generative freedom.

6.4 Casual Player Appeal

The findings of this study also showcased how open-ended dialogue tends to have an increased appeal towards casual players. A plausible interpretation is that casual players may value conversational expression and atmospheric immersion more highly than efficiency or mechanical optimization. In contrast, hardcore players may prioritize task progression, clarity, and predictability, which are qualities that generative systems often compromise.

7 Design Framework and Implications

Building on the empirical results and qualitative analysis from the Quest of Aivengarde study, this section synthesizes the findings into a design-oriented framework for LLM-driven NPC dialogue. Rather than treating dialogue systems as a binary choice between authored and generative approaches, the framework conceptualizes dialogue design as a set of trade-offs along two orthogonal dimensions: *authorial control versus conversational freedom*, and *system stability versus generative variability*. This framing is intended to support both researchers and practitioners in selecting and configuring dialogue systems that align with design goals, player audience, and technical constraints.

Figure 10 visualizes this design space and situates the four evaluated dialogue systems within it. The horizontal axis captures the degree of conversational freedom afforded to the player, ranging from tightly authored dialogue options to fully open-ended input. The vertical axis reflects system behavior, ranging from stable and predictable dialogue delivery to increasingly variable and emergent generative output.

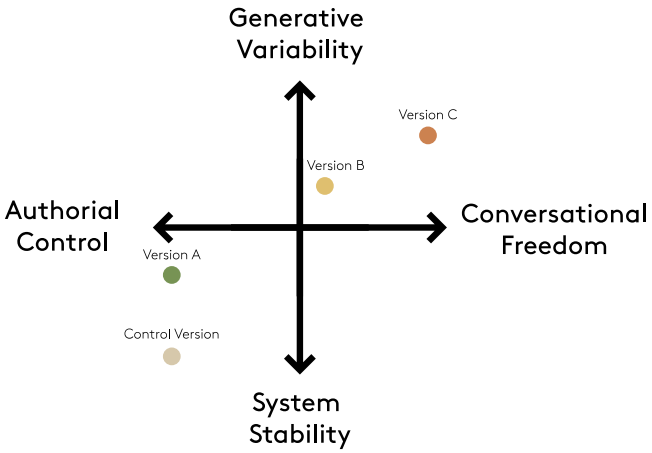


Figure 10: Design framework situating the four dialogue systems along axes of authorial control versus conversational freedom (horizontal) and system stability versus generative variability (vertical). Positions are conceptual and derived from empirical and qualitative analysis.

7.1 A Control–Freedom Design Space

As shown in Figure 10, the four dialogue systems span a continuum from fully authored to fully generative interaction. The horizontal axis reflects design intent. Systems positioned further to the left emphasize deterministic progression, predictable information delivery, and narrative consistency, while systems to the right prioritize expressive player input, conversational flexibility, and emergent

role-play. The vertical axis captures system behavior. Lower positions indicate reliable and repeatable outputs, while higher positions indicate increased linguistic variation, emergent responses, and corresponding risk of inconsistency or narrative drift.

The *Control Version* occupies the lower-left region of the space, characterized by high authorial control and high system stability. Version C appears in the upper-right region, reflecting high conversational freedom coupled with high generative variability. Versions A and B occupy intermediate positions, introducing generative elements while retaining differing degrees of structural constraint.

Importantly, movement along one axis often introduces trade-offs along the other. Increasing conversational freedom tends to amplify generative variability, while enforcing strict structural control typically improves stability. Dialogue design thus becomes a matter of informed compromise rather than optimization.

7.2 Appropriate Use Cases for Dialogue System Types

The empirical findings indicate that no single dialogue system is universally optimal. Instead, each occupies a distinct region of the design space and is well suited to particular design contexts.

Version A (LLM rephrasing) is positioned left of center with moderate generative variability (Figure 10). It is most appropriate when developers seek to reduce repetition and increase perceived freshness without altering gameplay logic or narrative structure. Because all dialogue content remains authored and outcomes are fixed, this approach preserves strong narrative control and system stability. However, results show that surface-level variation alone does not substantially enhance engagement or perceived agency.

Version B (hybrid fixed and open-ended dialogue) occupies a middle position along the generative variability axis while remaining relatively low on conversational freedom. This configuration is well suited for contexts where players benefit from clarification, follow-up questions, or expressive role-play, but where core narrative beats must remain tightly controlled. Empirically, this version was particularly valued by casual players, though qualitative feedback revealed sensitivity to response relevance and system consistency.

Version C (fully open-ended dialogue) appears in the upper-right region of the design space, reflecting high conversational freedom and high generative variability. *Version C* is most appropriate when the design goal prioritizes immersion, conversational naturalness, and player agency over strict predictability. The results show that this approach fosters longer and more engaging interactions without increasing non-dialogue gameplay time, especially for casual players. At the same time, it introduces higher risk in terms of inconsistency, narrative drift, and reliance on robust prompt design and control mechanisms.

7.3 Design Principles for LLM-Driven Dialogue

Across dialogue systems, several design principles emerge:

Balancing structure and freedom. Complete conversational freedom is not inherently superior. Players value expressive agency, but they also rely on NPCs to deliver accurate, timely, and relevant information. Systems that allow free input while maintaining hidden structural constraints, such as gated knowledge release or quest validation modules, can mitigate this tension.

Accounting for player type and context. Casual players in particular responded positively to open-ended and hybrid dialogue, emphasizing immersion and conversational naturalness. More experienced or goal-oriented players may place greater value on efficiency and clarity. Dialogue systems should therefore be selected with the target audience and play context in mind.

Respecting technical constraints. Generative dialogue introduces practical concerns including latency, cost, and moderation. Even modest increases in response time were noted by participants, and inconsistent outputs negatively affected trust in the system. Designers should treat LLM integration as a systems problem, where dialogue quality depends as much on orchestration, caching, and validation as on the model itself.

7.4 Applying the Framework

For developers, the framework provides a decision model for aligning dialogue systems with narrative ambition, production constraints, and player expectations. Rather than defaulting to fully open-ended interaction, teams can deliberately choose intermediate configurations that balance expressive freedom with stability.

For researchers, the framework offers a structured vocabulary for comparing dialogue systems beyond binary distinctions. By situating systems within a shared design space, future studies can more clearly articulate what aspects of dialogue design are being varied and evaluated, supporting cumulative and comparable research on LLM-driven interaction in games.

Overall, the framework reframes LLM-driven NPC dialogue not as a replacement for traditional systems, but as an expanded design space in which different configurations offer distinct experiential and technical trade-offs.

8 Conclusion and Future Work

8.1 Summary of Key Findings

This study explores how different large language models can be implemented and used to create dialogue systems for NPCs, and how these systems influence player interaction. Findings suggest that open-ended LLM dialogue enhances engagement and immersion. The results show differences across the different versions tested in this study, with *Version C* standing out both in the objective gameplay metrics and the subjective participant feedback.

Gameplay Time. Participants took significantly longer to complete *Version C* compared to the other versions. However, when dialogue time was subtracted, the remaining gameplay showed no difference. This suggests that the added duration was caused by longer and more extensive conversations. For casual players, the completion time was similar across all versions, although dialogue time was significantly higher in *Version C*.

Dialogue Interactions. The total amount of dialogue time was significantly higher in *Version C*, with a large effect size, indicating deeper engagement with the dialogue system. The *Control Version* resulted in more frequent interaction initiations, while *Version B* had

longer dialogues per exchange. *Version C* also showed significantly longer dialogue interactions with several NPCs.

Quest Completion. A majority of the quests showed no significant variation in completion time or failure rate. However, the quest *Break the Barrier* took significantly longer in *Version C*.

LLM Loading Times. Version C showed the shortest average loading time per response, but the highest total LLM loading time across a full playthrough.

Survey Feedback. Results showed that the participants preferred the NPCs in *Version C*, as they found them significantly more interesting to converse with compared to *Version A*. In addition, there was also a significant difference in the feeling of how enjoyable the system felt and in reports of moments where the dialogue felt unnatural or confusing.

Qualitative Feedback. The positive comments for *Version B* and *Version C* often mentioned immersion, agency, and flexibility. However, across all versions, the participants pointed out recurring issues such as limited NPC responsiveness, a tendency towards occasional misinformation, and user interface challenges. The suggestions for improvement largely focused on UI design, quest clarity, and dialogue functionality.

Player-Written Responses. In *Version B*, 18% of all players' responses were written freely, with most being short and in-character. In *Version C*, over 95% of the responses remained grounded in the game world, suggesting that the open-ended system supported creative yet contextually appropriate player immersion and role-play.

Sythesis in Relation to RQ1. In summary, the results provide a clear answer to RQ1. Dialogue systems that afford higher degrees of generative freedom substantially increase player engagement and immersion in narrative gameplay. Fully open-ended interaction in *Version C* consistently led to longer dialogue sessions, more sustained NPC conversations, and stronger subjective evaluations of enjoyment and interest. These findings suggest that the observed engagement gains stem not from altered task structure or progression, but from the conversational affordances themselves, which encourage players to treat NPCs as interactive social agents rather than purely functional quest providers.

Sythesis in Relation to RQ2. In addressing RQ2, the study reveals a set of systematic trade-offs between authorial control, system stability, and player agency. Systems with high authorial control and fixed structure, such as the *Control Version* and *Version A*, ensured predictable narrative progression and technical reliability, but limited expressive player agency and conversational depth. Conversely, the fully open-ended design of *Version C* maximised player agency and immersion at the cost of increased generative variability, including occasional inconsistencies and latency. The hybrid approach of *Version B* occupied an intermediate position, enabling expressive input while maintaining narrative constraints. However, it introduced its own coherence and usability challenges. These results indicate that generative dialogue design is not a question of superiority between authored and open-ended systems, but of informed compromise, where increasing conversational freedom inherently amplifies demands on system control.

8.2 Future Work and Recommendations

The results and insights from this study are promising. However several areas for future research and potential improvements should be considered. Given the central role of LLM integration in this study, it is important to look at the testing methodology, technical aspects, and design considerations that can be adjusted for further refinement. Future studies should include a broader and more diverse participant group to improve generalizability. It should include a more varied participant player base and add emphasis on including different backgrounds, as these perspectives might reveal new interaction patterns and player preferences. Additionally, including a within-subject test design would allow for players to compare multiple dialogue systems with the possibility of a more comparative study. This would also entail a restructuring of the design of the game to allow for replayability, so it does not impact subsequent playthroughs.

In terms of system design, LLM memory management is also subject to improvements that can benefit future work. To maintain conversational consistency over time, both long-term and short-term memory layers should be implemented. This would increase contextual narrative consistency with the possibility of yielding a more meaningful Player-NPC interaction. Other strategies, such as Chain-of-Thought prompting, could provide the LLM with improved reasoning skills. These improvements would overall increase the reasoning power and skill, resulting in more accurate LLM outputs.

This study offers practical insight when it comes to selecting LLM-powered dialogue systems. The more structured systems, such as *Version A*, are practical for usage scenarios where cost efficiency, narrative precision, and increased technical control are essential, such as games with a rigid narrative. This approach is affected by higher average LLM loading times and is viewed as less useful compared to other model designs. In contrast, more open-ended systems such as *Version B* and *Version C* provide the possibility of creative freedom and nuanced player engagement. For casual players, the free-form systems might feel more intuitive, but they can also be susceptible to unpredictable behaviors, such as hallucinations or inconsistencies.

These findings provide concrete principles for a path towards future design considerations regarding LLM integration in player-centered NPC dialogue systems. Designing such systems requires technical considerations and iterative design and improvements, but potentially results in LLM-powered dialogue systems with the potential to significantly enhance narrative design and player engagement in future game experiences.

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References

[1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).

[2] Nader Akoury, Qian Yang, and Mohit Iyyer. 2023. A framework for exploring player perceptions of llm-generated dialogue in commercial video games. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 2295–2311.

[3] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Deep Ganguli Goel, et al. 2022. Training a Helpful and Harmless Assistant with RLHF. *arXiv preprint arXiv:2204.05862* (2022).

[4] Gary Charness, Uri Gneezy, and Michael A Kuhn. 2012. Experimental methods: Between-subject and within-subject design. *Journal of economic behavior & organization* 81, 1 (2012), 1–8.

[5] Xinyun Chen, Jianfeng Gao, et al. 2024. Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone. *arXiv preprint arXiv:2404.14219* (2024).

[6] Jacob Cohen. 1988. *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge, Hillsdale, NJ.

[7] Chris Crawford. 2004. *Chris Crawford on interactive storytelling*. Pearson Education.

[8] Google DeepMind. 2023. Gemini: A Family of Highly Capable Multimodal Models. *arXiv preprint arXiv:2312.11805* (2023).

[9] Sebastian Domsch. 2017. Dialogue in video games. In *Dialogue across media*. John Benjamins Publishing Company, 251–270.

[10] Leon Tristan Dratzidis, Nima Zargham, and Rainer Malaka. 2025. "Follow My Lead": Role of AI-Based NPC Autonomy in Player-NPC Collaboration. In *International Conference on Entertainment Computing*. Springer, 60–77.

[11] Roberto Gallotta, Graham Todd, Marvin Zammit, Sam Earle, Antonios Liapis, Julian Togelius, and Georgios N Yannakakis. 2024. Large language models and games: A survey and roadmap. *IEEE Transactions on Games* (2024).

[12] Minh Hua and Rita Raley. 2020. Playing With Unicorns: AI Dungeon and Citizen NLP. *DHQ: Digital Humanities Quarterly* 14, 4 (2020).

[13] Junyang Huang. 2024. Generating dynamic and lifelike NPC dialogs in role-playing games using large language model. (2024).

[14] Jin Jeong and Tak Yeon Lee. 2025. LIGS: Developing an LLM-infused Game System for Emergent Narrative. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–12.

[15] Zexuan Ji, Nayeon Lee, Jason Fries, et al. 2023. A survey on hallucination in natural language generation. *ACM Computing Surveys (CSUR)* 55, 12 (2023), 1–38.

[16] Atsushi Kikumoto, Jason Hubbard, and Ulrich Mayr. 2016. Dynamics of task-set carry-over: evidence from eye-movement analyses. *Psychonomic bulletin & review* 23 (2016), 899–906.

[17] Mina Lee, Percy Liang, and Qian Yang. 2022. Coauthor: Designing a human-ai collaborative writing dataset for exploring language model capabilities. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–19.

[18] Chuyuan Li, Yuwei Yin, and Giuseppe Carenini. 2024. Dialogue Discourse Parsing as Generation: A Sequence-to-Sequence LLM-based Approach. In *Proceedings of the 25th Annual Meeting of the Special Interest Group on Discourse and Dialogue*. 1–14.

[19] Xiao Liu, Zhenping Xie, and Senlin Jiang. 2025. Personalized Non-Player Characters: A Framework for Character-Consistent Dialogue Generation. *AI* 6, 5 (2025), 93.

[20] Michael Mateas and Andrew Stern. 2003. Façade: An experiment in building a fully-realized interactive drama. In *Game developers conference*, Vol. 2. Citeseer, 4–8.

[21] Frans Mäyrä. 2017. Dialogue and interaction in role-playing games: Playful communication as Ludic culture. In *Dialogue across Media*. John Benjamins Publishing Company, 271–290.

[22] Gaël Mialon, Teven Le Scao, Xuechen Zhang, et al. 2023. Augmented language models: a survey. *arXiv preprint arXiv:2302.07842* (2023).

[23] Xiangyu Peng, Jessica Quaye, Sudha Rao, Weijia Xu, Portia Botchway, Chris Brockett, Nebojsa Jojic, Gabriel DesGarennes, Ken Lobb, Michael Xu, et al. 2024. Player-driven emergence in llm-driven game narrative. In *2024 IEEE Conference on Games (CoG)*. IEEE, 1–8.

[24] Rasmus Ploug, Emil Rimer, Anthon Kristian Skov Petersen, and Marco Scirea. 2025. Open-Ended NPC Dialogue Favors Casual Players: A Pilot Comparison of Three LLM-Driven Dialogue Systems. In *2025 IEEE Conference on Games (CoG)*. IEEE, 1–4.

[25] Christian Poglitsch, Fabian Szakács, and Johanna Pirker. 2025. Evaluating Large Language Models through Communication Games: An Agent-Based Framework Using Werewolf in Unity. In *Proceedings of the 20th International Conference on the Foundations of Digital Games*. 1–10.

[26] Mark Owen Riedl and Vadim Bulitko. 2013. Interactive narrative: An intelligent systems approach. *Ai Magazine* 34, 1 (2013), 67–67.

[27] Mark O Riedl and Robert Michael Young. 2010. Narrative planning: Balancing plot and character. *Journal of Artificial Intelligence Research* 39 (2010), 217–268.

[28] Emil Rimer, Rasmus Ploug, Anthon Kristian Skov Petersen, and Marco Scirea. 2025. Talking to NPCs: Three LLM-Driven Approaches to Dynamic RPG Dialogue. In *2025 IEEE Conference on Games (CoG)*. IEEE, 1–2.

[29] Li Song. 2025. LLM-Driven NPCs: Cross-Platform Dialogue System for Games and Social Platforms. *arXiv preprint arXiv:2504.13928* (2025).

[30] Penny Sweetser. 2024. Large language models and video games: A preliminary scoping review. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces*. 1–8.

[31] Leanne Taylor-Giles. 2020. Player-centred design in role-playing game branching dialogue Systems. *Game user experience and player-centered design* (2020), 295–325.

[32] Mike Treanor, Ben Samuel, and Mark J Nelson. 2025. Slice of Life: A Social Physics Game with Interactive Conversations using Symbolically Grounded LLM-Based Generative Dialogue. In *Proceedings of the 20th International Conference on the Foundations of Digital Games*. 1–11.

[33] Henrik Warpefelt. 2016. *The Non-Player Character: Exploring the believability of NPC presentation and behavior*. Ph. D. Dissertation. Department of Computer and Systems Sciences, Stockholm University.

[34] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.

[35] Jim Whitehead, Thomas Wessel, Blythe Chen, Raven Cruz-James, Luc Harnist, William Klunder, Justin Lam, Ethan Lin, Roman Luo, Hung Nguyen, et al. 2025. Conversational Interactions with Procedural Generators using Large Language Models. In *Proceedings of the 20th International Conference on the Foundations of Digital Games*. 1–7.

[36] Yuhang Wu, Weijie Yin, Zhaoyang Yang, Jianzong Jiang, Zhenhua Yuan, Chin-Yew Lin, et al. 2023. Visual ChatGPT: Talking, Drawing and Editing with Visual Foundation Models. *arXiv preprint arXiv:2303.04671* (2023).

[37] Daijin Yang, Erica Kleinman, and Casper Harteveld. 2024. GPT for games: A scoping review (2020-2023). In *2024 IEEE Conference on Games (CoG)*. IEEE, 1–8.

[38] Nima Zargham, Aaron Merkord, Ameneh Safari, Leon Tristan Dratzidis, and Rainer Malaka. 2025. "Can You Feel Me Now?": Exploring Player Empathy in AI-Based NPC Conversations. In *International Conference on Entertainment Computing*. Springer, 389–407.

[39] Shijie Zheng, Keith He, Le Yang, and Jie Xiong. 2024. MemoryRepository for AI NPC. *IEEE Access* (2024).

[40] Andrew Zhu, Lara J. Martin, Andrew Head, and Chris Callison-Burch. 2023. CALYPSO: LLMs as Dungeon Masters' Assistants. In *Proceedings of the 19th AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*.